

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388448
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Shen; Kuan-Yueh

Phase lock loop with an adaptive loop filter

Abstract

An apparatus has a phase lock loop with an adaptive loop filter that has a reset circuit controlled by a power gating pulse circuit.

Inventors:	Shen; Kuan-Yueh (Portland, OR)
Applicant:	Intel Corporation (Santa Clara, CA)
Family ID:	82899030
Assignee:	Intel Corporation (Santa Clara, CA)
Appl. No.:	17/482015
Filed:	September 22, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230092886 A1	Mar. 23, 2023

Publication Classification

Int. Cl.: H03L7/093 (20060101); H03L7/08 (20060101); H03L7/089 (20060101)

U.S. Cl.:

CPC H03L7/093 (20130101); H03L7/0802 (20130101); H03L7/089 (20130101);

Field of Classification Search

CPC: H03L (7/0802); H03L (7/089); H03L (7/093); H03L (7/107); H03L (7/1075)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8901994	12/2013	Fan	N/A	N/A
10574243	12/2019	Shen et al.	N/A	N/A
10700688	12/2019	Fan et al.	N/A	N/A
2018/0191359	12/2017	Petrov	N/A	H03L 7/0891
2020/0195255	12/2019	Fan et al.	N/A	N/A

OTHER PUBLICATIONS

Maxim, Adrian et al., “A Low-Jitter 125-1250-MHz Process-Independent and Ripple-Poleless 0.18-um CMOS PLL Based on a Sample-Reset Loop Filter” , IEEE Journal of Solid-State Circuits, vol. 36, No. 11, Nov. 2001, 11 pgs. cited by applicant

Shen, Kuan-Yueh et al., “A Flexible, Low-Power Analog PLL for SoC and Processors in 14nm CMOS”, IEEE Transactions on Circuits and Systems—I: Regular Papers, vol. 65, No. 7, Jul. 2018, 9 pgs. cited by applicant

Shen, Kuan-Yueh James et al., “A 0.17-to-3.5mW 0.15-to-5GHZ SOC PLL With 15dB Built-In Supply Noise Rejection and Self-Bandwidth Control in 14nm CMOS” , ISSCC 2016, Session 19, Digital PLLs, 19.4, 3 pgs. cited by applicant

European Search Report and Search Opinion, EP App. No. 22190037.6, Feb. 20, 2023, 12 pages. cited by applicant

Lim, P. J., “An Area-Efficient PLL Architecture in 90-nm CMOS”, Symposium on VLSI Circuits Digest of Technical Papers, Jun. 16, 2005, pp. 48-49. cited by applicant

Office Action, EP App. No. 22190037.6, Jul. 3, 2025, 6 pages. cited by applicant

Primary Examiner: Neill; Patrick O

Attorney, Agent or Firm: NICHOLSON DE VOS WEBSTER & ELLIOTT LLP

Background/Summary

BACKGROUND

(1) A phase locked (or lock) loop (PLL) is a clock control system to provide an output signal with a phase related to the phase of the input signal, often used to control an oscillator. One conventional PLL configuration uses a sample-reset loop filter (SR-LF) that resets capacitors on the loop filter circuit to avoid memory effect that could undesirably reduce the PLL phase margin or stability causing higher variation of PLL lock time, or more frequency ripples in a PLL phase step response. The known reset circuits used by the SR-LF, however, often are very inefficient consuming much more power than necessary, and overly expensive due to the type of components (e.g., precision resistors which may add significant manufacturing process costs) on the circuit as well as the typical required footprint size needed for the reset circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The implementations of the disclosure will be understood more fully from the detailed

description given below and from the accompanying drawings of various implementations of the disclosure, which, however, should not be taken to limit the disclosure to the specific implementations, but are for explanation and understanding only.

(2) FIG. 1 is a timing diagram of a conventional SR-LF for a phase lock loop;

(3) FIG. 2 is a schematic diagram of an example phase lock loop according to at least one of the implementations herein;

(4) FIG. 3 is another diagram of an example phase lock loop according to at least one of the implementations herein;

(5) FIG. 4 is a diagram of an example phase frequency detector of a phase lock loop according to at least one of the implementations herein;

(6) FIG. 5 is a diagram of an example power gating pulse circuit according to at least one of the implementations herein;

(7) FIG. 6 is a timing diagram of an example phase lock loop according to at least one of the implementations herein;

(8) FIG. 7 is a diagram of an alternative example power gating signal circuit according to at least one of the implementations herein;

(9) FIG. 8 is a diagram of an example loop filter according to at least one of the implementations herein;

(10) FIG. 9A is a diagram of an example loop filter reset circuit for proportional control according to at least one of the implementations herein;

(11) FIG. 9B is a diagram of an example alternative loop filter reset circuit for proportional control according to at least one of the implementations herein; and

(12) FIG. 10 is a schematic diagram of a system according to at least one of the implementations herein.

DETAILED DESCRIPTION

(13) Embodiments discussed herein variously provide techniques and mechanisms for regulating operation of a PLL circuit. In the following description, numerous details are discussed to provide a more thorough explanation of the embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring embodiments of the present disclosure.

(14) Note that in the corresponding drawings of the embodiments, signals are represented with lines. Some lines may be thicker, to indicate a greater number of constituent signal paths, and/or have arrows at one or more ends, to indicate a direction of information flow. Such indications are not intended to be limiting. Rather, the lines are used in connection with one or more exemplary embodiments to facilitate easier understanding of a circuit or a logical unit. Any represented signal, as dictated by design needs or preferences, may actually comprise one or more signals that may travel in either direction and may be implemented with any suitable type of signal scheme.

(15) Throughout the specification, and in the claims, the term “connected” refers to a direct connection, such as electrical, mechanical, or magnetic connection between the things that are connected, without any intermediary devices. The term “coupled” refers to a direct or indirect connection, such as a direct electrical, mechanical, or magnetic connection between the things that are connected or an indirect connection, through one or more passive or active intermediary devices. The term “circuit” or “module” may refer to one or more passive and/or active components that are arranged to cooperate with one another to provide a desired function. The term “signal” may refer to at least one current signal, voltage signal, magnetic signal, or data/clock signal. The meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

(16) The term “device” may generally refer to an apparatus according to the context of the usage of

that term. For example, a device may refer to a stack of layers or structures, a single structure or layer, a connection of various structures having active and/or passive elements, etc. Generally, a device is a three-dimensional structure with a plane along the x-y direction and a height along the z direction of an x-y-z Cartesian coordinate system. The plane of the device may also be the plane of an apparatus which comprises the device.

(17) The term “scaling” generally refers to converting a design (schematic and layout) from one process technology to another process technology and subsequently being reduced in layout area. The term “scaling” generally also refers to downsizing layout and devices within the same technology node. The term “scaling” may also refer to adjusting (e.g., slowing down or speeding up—i.e. scaling down, or scaling up respectively) of a signal frequency relative to another parameter, for example, power supply level.

(18) The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within $\pm 10\%$ of a target value. For example, unless otherwise specified in the explicit context of their use, the terms “substantially equal,” “about equal” and “approximately equal” mean that there is no more than incidental variation between among things so described. In the art, such variation is typically no more than $\pm 10\%$ of a predetermined target value.

(19) It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(20) Unless otherwise specified the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

(21) The terms “left,” “right,” “front,” “back,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. For example, the terms “over,” “under,” “front side,” “back side,” “top,” “bottom,” “over,” “under,” and “on” as used herein refer to a relative position of one component, structure, or material with respect to other referenced components, structures or materials within a device, where such physical relationships are noteworthy. These terms may be employed herein for descriptive purposes only and predominantly within the context of a device z-axis and therefore may be relative to an orientation of a device. Hence, a first material “over” a second material in the context of a figure provided herein may also be “under” the second material if the device is oriented upside-down relative to the context of the figure provided. In the context of materials, one material disposed over or under another may be directly in contact or may have one or more intervening materials. Moreover, one material disposed between two materials may be directly in contact with the two layers or may have one or more intervening layers. In contrast, a first material “on” a second material is in direct contact with that second material. Similar distinctions are to be made in the context of component assemblies.

(22) The term “between” may be employed in the context of the z-axis, x-axis or y-axis of a device. A material that is between two other materials may be in contact with one or both of those materials, or it may be separated from both of the other two materials by one or more intervening materials. A material “between” two other materials may therefore be in contact with either of the other two materials, or it may be coupled to the other two materials through an intervening material. A device that is between two other devices may be directly connected to one or both of those devices, or it may be separated from both of the other two devices by one or more intervening devices.

(23) As used throughout this description, and in the claims, a list of items joined by the term “at least one of” or “one or more of” can mean any combination of the listed terms. For example, the phrase “at least one of A, B or C” can mean A; B; C; A and B; A and C; B and C; or A, B and C. It is pointed out that those elements of a figure having the same reference numbers (or names) as the

elements of any other figure can operate or function in any manner similar to that described, but are not limited to such.

(24) In addition, the various elements of combinatorial logic and sequential logic discussed in the present disclosure may pertain both to physical structures (such as AND gates, OR gates, or XOR gates), or to synthesized or otherwise optimized collections of devices implementing the logical structures that are Boolean equivalents of the logic under discussion.

(25) With recent advancements in System-on-Chip (SoC) integration, modern SoC architectures can employ many PLLs (e.g., more than 20 PLLs). However, SoC clocking has an ever reducing power budget, and conventional PLLs are known to consume too much power. Specifically, conventional semiconductor chip PLLs, such as phase lock loops that provide low jitter and receive a reference signal and feedback signal. A phase frequency detector (PFD) detects a phase difference or error between the signals and provides UP and DOWN (DN) signals to indicate the phase difference. Charge pumps may receive the UP and DN signals to generate control voltage provided to the loop filter.

(26) Sample-reset loop filters (SR-LFs) have an integral path and a proportional path. The proportional path uses charge that is proportional to the phase difference between the reference and feedback signals (or clocks). A dual charge pump architecture in PLLs with low jitter highly decouples the proportional and integral control. The proportional control resets charges at capacitors in the proportional path in the SR-LF to erase memory effect per reference cycle. If the reset is incomplete due to insufficient reset current or reset duration where the capacitors do not return to their initial voltage in a reference cycle, the PLL does not reach a steady state and will need more reference cycles to reach the steady state (or achieve a longer lock time).

(27) Older SR-LF PLLs use unity gain buffer (UGB) reset circuits with dynamic input and output range and that typically resets two alternating capacitors. The UGB reset circuits have more recently been replaced and simplified by reset circuits that use resistors or voltage dividers and improve upon the UGB in terms of reduced power consumption, as well as a smaller and less complex circuit. For example, the resistor divider reset circuit power gates the high reset current so that the resistor divider's ON time is duty cycled down to about half or 51% of the reference cycle before a next charge pump sampling window opens, which saves current relative to the UGB reset circuit, but is still considered wasteful.

(28) Referring to FIG. 1 for example, a time chart **100** shows the conventional reset timing during SR-LF operation of a PLL. The PLL reference and feedback signals or clocks are shown with frequencies $F_{\text{sub.ref}}$ and $F_{\text{sub.tbk}}$ respectively. A net up-dn signal on time chart **100** represents the net phase difference or phase error pulse where the UP signal is the rising edge of a leading pulse of one of the reference and feedback signals, here being the reference signal, and where the DN signal of the net up-dn signal is the rising edge of the trailing pulse, here being the feedback pulse. The UP and DN signals are generated by a phase frequency detector (PFD) of the PLL, and the up-dn signal represents an XOR(up, dn) operation (only for the $F_{\text{sub.ref}}$ & $F_{\text{sub.tbk}}$ rising edges) where a pulse width $t_{\text{sub.p}}$ of the net up-dn signal represents a phase error or difference.

(29) Clocks cp1 and cp2 are charge pump sampling clocks that open and close sampling switches on a loop filter of the PLL to show the timing to provide charge to a proportional path of the loop filter. Clocks cp1a and cp2a are reset clocks that activate a reset circuit of the loop filter to open and close reset switches that provide current to two capacitors on the proportional path. All four of these clocks are conventionally derived from the PLL reference clock, and all run at $F_{\text{sub.ref}}/2$ in this case. Time at 0, as well as at t^* , begins the present and subsequent charge pump events, which refers to pumping charge to both of the integral and proportional paths of the loop filter.

(30) The timing chart **100** also shows the alternating operation to provide current to the two parallel lines forming the proportional path so that while one line receives pumped charge to provide a control signal, the other line performs a capacitive reset to voltage $V_{\text{sub.RST}}$ in some configurations. Most relevant here, the reset durations show, by the pulse widths on the reset clocks

cp1a and cp2a, that the resets are conventionally set to be sustained for about half of the reference cycle. The reset duration, however, only needs to fully reset nodes (on the parallel lines interconnecting the parallel lines to the reset switches) back to V.sub.RST before the reset switches disconnect, respectively. Thus, the extra resetting time wastes power with the high pulse on cp1a and cp2a that each lasts about 25% of the reference cycle.

(31) In addition, given the reference frequency F.sub.ref, the PLL changes its feedback divide ratio N to support dynamic frequency scaling accomplished by using a frequency divider on the feedback of the PLL. If the charge pump current can be linearly proportional to N, the PLL maintains its loop stability and bandwidth. In such a configuration, the higher the N ratio, the more charge is pumped to the capacitors of the proportional path (assuming similar PLL phase errors, or up-dn pulse widths, across N's), thus requiring a higher reset current to fully reset the proportional path capacitors across the range of possible values for the N ratios. Since the high N scenario decides the high reset current for N of any value, for a low N ratio (such as down to four), the high reset current wastes more power.

(32) Specifically, when a PLL is controlling a voltage controlled oscillator (VCO), the output frequency F.sub.vco (from the VCO) can trend between about 400 MHz to 6+ GHz. In this case, the same reset current in the SR-LF is adequate at high frequencies toward the 6 GHz, but when the output frequency F.sub.vco is low towards the 400 MHz and the reset current is kept the same, then the reset current is way too excessive for the lower frequency output F.sub.vco, thereby wasting a relatively large amount of power. However, the reset current also cannot remain at a fixed low level because if the reset current is tuned too low, the PLL is known to have stability problems at the high output frequencies F.sub.vco. In order to conventionally avoid this latter difficulty then, discrete reset current levels are increased, thereby wasting power for PLL stability at high output frequencies.

(33) More specifically, a charge pump-based PLL at a minimum output frequency F.sub.min of 0.4 GHz with F.sub.ref at 100 MHz has SR-LF power consumption of about 220 μ A at a nominal corner, or 39% of the total PLL power from the PLL supply voltage V.sub.ccpll. In addition to the reasons mentioned above, this power drain should be reduced since additional SR-LF power does not increase PLL performance, such as by lowering PLL output phase noise or jitter.

(34) Thus, such conventional PLL devices have loop filters with fixed reset current that is not adaptive to changes in phase error, output frequencies, and charge pump strength, which wastes a large amount of power relative to certain types of circuits such as those on semiconductor chips.

(35) To resolve these issues, some implementations described herein are realized by a power gating pulse circuit that provides reset control signals to a reset circuit of a sample-reset loop filter in order to reset capacitors on the proportional path of the loop filter. The reset control signals have significantly shorter pulse widths as little as 10% or less of the reference signal or clock, and even down to about 3-4%. Since the reset time and power consumed by the reset circuit is based on pulse widths of reset control signals that adapt to the PLL phase error, this significantly reduces power consumption by the loop filter, and in turn, the phase locked loop (PLL) using the loop filter.

(36) By some implementations, this is accomplished by an apparatus, system, or device with a loop filter to receive input and to output a control signal based on the input. The loop filter comprises capacitors coupled to the input and a reset circuit to reset the capacitors. A power gating pulse circuit comprises pulse generator circuitry (whether a flip-flop or equivalent circuitry) configured to adaptively generate complimentary reset control signals to control the reset circuit, where a pulse width of the reset control signals is based on both (1) a phase error between a reference clock and a feedback clock associated with the input to the loop filter and (2) switch control signals based on the reference clock. This arrangement effectively modifies or reduces the pulse widths of those switch control signals.

(37) By some implementations, a phase frequency detection (PFD) reset signal is based on the detection of the phase error by the PFD, and the available PFD reset signal is provided to a power

gating pulse circuit. The power gating pulse (PGP) circuit may have a delay circuit to delay the PFD reset signal to generate a PGP (or flip-flop) reset signal. The PGP circuit also may have flip-flop circuitry with a flip-flop (or by one example, a D-type flip-flop) that is reset by the flip-flop reset signal to generate the termination of the pulses of the loop filter reset control signals. The flip-flop also may receive input from a Boolean circuit that converts switch control signals to the input of the flip-flop to generate the start of the pulses of the reset control signals. The switch control signals are the reset clocks that have pulse widths that are a fixed proportion of the reference clock. (38) By other implementations, the reset circuit is an active element voltage divider that uses at least, or exactly, four transistors, rather than resistors, to receive the reset control signals. (it doesn't provide the signals to other blocks.)

(39) These arrangements significantly reduce the reset duration at the loop filter as controlled by the reset control pulses. Also, the reset current can be realized by using a certain arrangement of the active devices in the reset circuit rather than using the resistor divider, which actually may increase the current while further reducing the reset duration, and still save power. This saves SR-LF power without increasing PLL jitter (or noise) or lock time. The reset configuration reduces the power consumed despite operation of a frequency divider for example. Also, the reset ON time becomes adaptive to the phase error, or in other words, a PFD up-dn net pulse width. For instance, if the phase error is larger at lower N with a lower charge pump current, the reset ON time tracks it automatically. In one form, the disclosed PLL may save up to about 69% of the loop filter power, and up to about 27% of the PLL power at a minimum frequency of 400 MHz (and from supply current $I_{sub.vccpll}$) at a nominal corner with a 100 MHz reference.

(40) Referring to FIG. 2, a device or system **200** may be a phase locked (or lock) loop) PLL and the implementations described herein treat the device **200** as a PLL. The PLL described herein may be a type I or type II PLL with a sample-reset loop filter (SR-LF) **206** with a proportional path, but could be other types of PLLs, or the device **200** even could alternatively be a delay locked loop (DLL), as long as a charge pump provides proportional control and resets are provided per reference cycle.

(41) For the present example, PLL **200** may have a phase frequency detector (PFD) **202**, one or more charge pumps **204**, the sample-reset loop filter (SR-LF) **206** with a reset circuit **214**, an integral path **216**, and a proportional path **218**, and a device being controlled or regulated (for frequency and/or phase) by the PLL such as a voltage control oscillator (VCO) **208**. The PLL also has a power gating pulse (PGP) circuit **212** to generate reset control signals with particular pulse widths as described herein. A feedback loop of the PLL may have a divider **220** to provide feedback with a variety of frequencies.

(42) The PFD **202** receives a reference clock signal or just reference clock RefClk with a frequency of $F_{sub.ref}$. The reference clock may be generated by any adequate known clock generation circuitry, such as a crystal oscillator, or another PLL (forming PLLs in cascade). A feedback signal or clock FbClk with a frequency $F_{sub.tbk}$ also is received to determine the phase difference, if any, between the feedback and reference frequencies. The feedback signal is looped from the output of a device being controlled by the PLL, such as the voltage controlled oscillator (VCO) **208**.

(43) The PFD **202** has circuitry to detect a phase difference or error between the reference and feedback signals, and forms an UP signal and a down DN signal. The UP and DN signals are then provided to a charge pump circuit **204**. The charge pump circuit **204** provides an integral path current $I_{sub.cpi}$ to the integral path **216** of the SR-LF **206** and a proportional current $I_{sub.cpp}$ to the proportional path **218**, where charging and/or discharging time utilizing $I_{sub.cpi}$ and $I_{sub.cpp}$ is proportional to the phase error.

(44) The power gating pulse (PGP) circuit **212** receives the PFD reset signal from the PFD circuit **202** as well as reset switch control signals cp1a and cp2a generated within the SR-LF **206** by sourcing and using the reference clock RefClk, for example. The reset switch control signals cp1a and cp2a are derived from RefClk, as disclosed by Maxim, A., et al., "A Low-Jitter 125-1250 Mhz

Process Independent and Ripple-Poleless 0.18 μm CMOS PLL Based on a Sample-Reset Loop Filter, IEEE Journal of Solid-State Circuits, Vol. 36, No. 11, (November 2001), which is incorporated herein for all purposes.

(45) The PGP circuit outputs reset control signals (or power gating (PG/PG') signals, and provides them to the reset circuit **214**. The reset circuit **214** resets the energy or charge storage elements or here, capacitors, on the proportional path **218** depending on the pulse widths of the PG and PG' signals. The SR-LF then provides a control signal V.sub.ctl to adjust the frequency of the VCO **208**.

(46) The divider **220** divides the clock or signal output from the VCO **208**, and divided down in frequency (e.g., divide by ratio 'N'). Other parallel feedback lines could be used with different divider ratio values (but only one of the feedback lines returns to PFD **202** as the FbClk). In some implementations, the divider may be an integer divider or a fractional divider. The divider can be implemented using any suitable divider circuitry. For example, Johnson Counters can be used for implementing dividers.

(47) Referring to FIG. 3, an apparatus, system, or device **300** provides more detail of the relevant circuits herein, and may be a PLL the same or similar to system **200**. Here, the system **300** has a PFD, or PFD circuit, **310**, charge pump circuitry **320**, a loop filter **330**, and a PGP circuit **340** (also referred to as a rapid reset circuit), all arranged as shown on FIG. 3. The PFD **310** receives the reference clock and feedback clock signals, which may be via one or more clock buffers to provide a delay when desired, and has an UP-DN circuit **312** that generates UP and DN signals **314** and **315** provided to charge pump circuitry **320**. The charge pump circuitry **320** generates path control current which is provided to the loop filter **330**. The loop filter **330** may have switches **332** including sample switches allowing charges from the charge pumps to a path on the loop filter and reset switches to erase the charges on capacitors on that path. The sample switches are closed depending on switch control signals **364** and **366** and are derived from the reference clock delivered to PFD **310**. A reset circuit **334** controls the reset switches for the loop filter to output a control signal V.sub.ctl to drive a device being controlled, such as the VCO.

(48) In order to provide the complimentary reset control signals PG/PG' **356** and **358** to the reset circuit **334**, the PFD **310** also has the UP-DN circuit **312** that generates a PFD reset signal **316** to reset the UP-DN circuit **312** according to the phase error indicated by the UP and DN signals, and that is provided to a delay circuit **342** of the PGP circuit **340**. Whether the delay of the delay circuit **342** is predetermined manually and fixed, or a state machine or machine learning (ML) unit **344** is able to adjust the delay, the delay circuit **342** applies the delay to the PFD reset signal **316** to generate a PGP (or flip-flop) reset signal **346**.

(49) The PGP reset signal **346** is provided to timer circuitry **350** that has a Boolean logic circuit **352** and a pulse generator circuit **354**, which may be a flip-flop circuit that has one or more D-type flip-flop (D-FF)s. The Boolean logic circuit **352** receives reset switch control signals **360** and **362** from the loop filter **330** to set the start or rise of the reset control pulses PG and PG' **356** and **358**. The Boolean logic circuit **352** generates a signal **353**, such as a flip-flop input signal, received by the pulse generator circuit **354**. When the pulse generator circuit **354** is a flip-flop circuit, the PGP reset signal **346** resets the flip-flop to generate the fall of the pulses of the reset control signal **356** and complimentary rise of reset control signal **358**, and the reset control signals **356** and **358** are then provided to the reset control circuit **334**. The pulse generator circuit **354** also may have an optional chopper branch or circuit to adjust or limit the pulse widths of the reset control signals when the phase error is longer than a certain threshold, for example half of a reference period. Those elements of FIG. 3 having the same reference numbers (or names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such. More details of system **300** are provided below.

(50) Referring to FIG. 4, an example PFD **400** has an UP-DN circuit **402**, similar or the same as PFD **310** and UP-DN circuit **312**, and may have any suitable circuitry. The example UP-DN circuit **402** here has at least a first or reference flip-flop **404**, a second or feedback flip-flop **406**, an AND

gate **408**, a PFD delay circuit (or internal delay) **410**, a reset node **412**, and clock inputs **414** and **416** to receive the reference signal refclk and feedback signal fbkclk as arranged as shown on FIG. 4. The reference flip-flop **404** and feedback flip-flop **406** respectively receive the reference signal refclk and feedback signal fbkclk at clock inputs of the flip-flops **404** and **406**. The flip-flops **404** and **406** may be D-type flip-flops where the D input is set at HIGH. The flip-flop **404** may have an output $Q_{\text{sub.A}}$ to provide the UP signal or pulse, and a reset coupled to node **412**, while flip-flop **406** may have an output $Q_{\text{sub.B}}$ to provide the DN signal or pulse, and also has a reset coupled to node **412**. The UP and DN signals are provided to the charge pump circuitry **320**.

(51) Referring again to FIG. 3, the charge pump circuitry **320** may have any suitable circuitry known to use the UP and DN signals from the PFD **310** (or **400**) to provide charge to the loop filter **330**. By one form, the charge pump circuitry **320** may have dual charge pumps with an integral charge pump and a proportional charge pump that both receive the UP and DN signals and output charge pump positive or negative current depending on whether a received signal is the UP signal or DN signal. The charge pump circuitry **320** outputs one or more currents to loop filter **330**, for example, wherein a current **322** at a control voltage V_{ctl} is provided to one or more paths on the loop filter **330** as described below. The proportional, or the second, charge pump charges the voltage sample and reset node of the loop filter and also the output node of the second charge pump. The effect is that a charge pump charges or discharges at the node V_{ctl} (FIG. 8) according to the UP and DN pulses from the PFD. A given conductance path of loop filter **330** accumulates charges proportional to the phase error whose sign is subject to whether UP or DN is leading and whose magnitude follows the pulse width of UP-DN. In some embodiments, a proportional path accepts charges like a corresponding integral path, but with usually a larger amount which is accumulated by a capacitor that must be discharged back or reset to its initial state before the next UP and DN event. Herein, the present SR-LF efficiently resets the charges in the proportional path.

(52) Returning again to FIG. 4, the AND gate **408** of the UP-DN circuit **402** receives both UP and DN (or $Q_{\text{sub.A}}$ and $Q_{\text{sub.B}}$) signals or pulses. The UP and DN signals may have the same frequency but are out of phase from each other where the difference in rising edges is the phase error $t_{\text{sub.p}}$ between the reference and feedback signals. The AND gate **408** primes when both UP and DN are high. The duration of that is controlled by the delay circuit or line T_x **410**. T_x should be long enough to avoid dead zone issues.

(53) Specifically, a phase difference end (pde) at output of the AND **408** is then provided to the PFD internal delay circuit **410** to add internal PFD delay T_x to form a residual pulse width. The delay circuit **410** may be formed of a delay line of a series of invertors or buffers by one example, although other configurations could be used. The delay circuit **410** effectively decides the pulse width of the residual pulse, or in other words, whichever is the narrower pulse between pulse $Q_{\text{sub.A}}$ and $Q_{\text{sub.B}}$, which then may be considered as residual pulse $Q_{\text{sub.A}}$ or residual pulse $Q_{\text{sub.B}}$. More precisely, a significant fraction of the narrower pulse or residual pulse is formed by T_x by adding delay to the pde signal. The residual pulse is provided at node X **412** to be provided to the flip-flops **404** and **406** to perform the PFD resets, so that the residual signal or pulse also may be considered the PFD reset pulse or signal. The residual pulse also may be referred to as the X pulse or signal. The PFD reset signal X may be generated once every reference cycle. The PFD reset signal X is then provided to the PGP circuit **340** to represent the end of the phase difference between the reference and feedback signals.

(54) Note that labels for nodes and signals on those nodes are interchangeably used. For example, $V_{\text{sub.ctl}}$ may refer to voltage $V_{\text{sub.ctl}}$ or node $V_{\text{sub.ctl}}$ depending on the context of the sentence.

(55) Referring to FIG. 5, an example PGP circuit (or rapid reset circuit) **500** is similar or the same as the PGP circuit **340**, and receives the PFD reset signal X and switch control signals cp1a and cp2a to generate reset control signals PG and PG'. Those elements of FIG. 5 having the same reference numbers (or similar or same names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such. The PGP circuit **500**

may have a delay circuit **502** as well as a Boolean logic circuit **504** and a pulse generator circuit (or flip-flop circuit) **506** to form the timer circuitry **350** (FIG. 3 for example).

(56) The delay circuit **502** receives the PFD reset signal X and applies a delay $t_{sub.D}$, to generate the PGP reset signal, or flip-flop reset signal, PGP.sub.rst. The delay circuit **502** here may be a delay line with a series of invertors or buffers where each invertor adds a stage that causes a certain amount of delay, but other equivalent configurations may be used instead. The delay $t_{sub.D}$ may be a tunable delay. Specifically, the number of delay stages may be programmable. In other words, the delay can be predetermined and tuned manually by selecting how many elements (invertors) or buffer stages (or delay stages) are to be active in series in the delay line in order to set a fixed delay duration. Here, about 200 ps may be used as the $t_{sub.D}$ target. The number of stages, however, still can vary depending on the process skew corners as well as the specified supply voltage, and therefore the system may heuristically derive the correct number of stages.

(57) By another form, the delay circuit **502** may be automatically tunable and a machine learning unit **344** (FIG. 3) may use a state machine to adapt the delay $t_{sub.D}$ as is necessary. This may include having the state machine automatically monitor the PG pulse width by a background feedback loop. Alternatively, one or more pre-trained neural networks may receive phase error signal data as input and then output a desired time delay $t_{sub.D}$ to adjust the PG pulse widths. Many variations are contemplated.

(58) Referring to FIG. 6, the Boolean logic circuit **504** receives reset switch control signals $cp1a$ and $cp2a$. As shown on the timing diagram **600** (FIG. 6), the reset switch control signals $cp1a$ and $cp2a$ may be the conventional reset control signals and are a proportion of the reference clock shown at frequency $F_{sub.ref}$. $Cp1$, $cp1a$, $cp2$, and $cp2a$ are non-overlapping clocks all with a period of $\frac{1}{2}F_{ref}$ [2 times the period of F_{ref}]. The switch control signals $cp1a$ and $cp2a$ alternate every reference cycle ($cp1a.fwdarw.cp2a.fwdarw.cp1a.fwdarw. \dots$) where $cp1a$ or $cp2a$ is HIGH for one-half a reference cycle. Together signals $cp1a$ and $cp2a$ have a pulse width about 50% of the reference cycle. The feedback clock $F_{sub.tbk}$ is shown to trail the reference clock at a phase difference or phase error $t_{sub.p}$. A net UP-DN pulse also is shown that corresponds to the phase error and is the difference in the rising edges of the UP pulse ($Q_{sub.A}$) and DN pulse ($Q_{sub.B}$) generated by the PFD **400**. Sampling switch control signals $cp1$ and $cp2$, which maybe generated by the SR-LF or PFD, and not necessarily the charge pumps, also are shown with alternating pulses to show the alternating operation of capacitors $C2a$ and $C2b$ on the proportion path of the loop filter **330**, where while one capacitor, say $C2a$, is receiving charge (from I_{cpp}), the other capacitor $C2b$ is being reset.

(59) Referring again to FIG. 5, the Boolean circuit **504** may have an OR gate **508** to receive the reset switch control signals $cp1a$ and $cp2a$ and output a signal Y that has a rising edge that indicates the rise of either switch control signal, and the Y rising edge occurs once every reference cycle. Two or more buffers in series **510** then may delay the Y signal to generate a signal Y1 in order to adjust the timing to flop HIGH from D to Q of a D-type flip-flop (D-FF) **512**. The delay of Y1 relative to Y postpones the onset time of the proportional charge reset process. When the delay is optimal, voltage ripple due to resetting, say capacitor $C2a$, may cancel the voltage ripple due to charging capacitor $C2b$. Such cancellation leads to low perturbation coupling onto V_{ctl} , resulting in lower VCO frequency fluctuation or deterministic jitter.

(60) On the pulse generating circuit **506**, the D-type flip-flop **512** has a reset input that resets output Q to LOW. The D input is held at HIGH. The flip-flop **512** has a reset that is activated by the PGP reset signal PGP.sub.rst. The output Q of the flip-flop **512** is the reset control pulse PG that is to be provided to the reset circuit **334** of the loop filter **330**. It will be appreciated that in alternative forms, the flip-flop could be a different type, or the D-type flip-flop could be replaced by components that perform equivalent operations. The reset control signal PG then may be inverted by an inverter **514** to also provide a complimentary reset control signal PG' to the reset circuit **334**.

(61) Referring again to FIG. 6, the timing diagram **600** shows the reset power gating by the reset

control pulses. A plot of signal Y (and in turn representing Y1) is shown, where Y is HIGH whenever cp1a or cp2a is HIGH. A plot of the PGP reset signal PGP.sub.rst also is shown. As indicated, Y sets the rising edge of the reset control pulses PG, while the PGP reset signal PGP.sub.rst sets the falling edge of the reset control pulse PG. As mentioned above with clock buffers, the reset control pulse PG actually may have its rising edge start very slightly before the charge pump event and therefore, before the rise of the UP-DN net signal. Otherwise, the charge pump event and the rise of the loop filter reset can happen almost simultaneously to help cancel the coupling from each other and onto Vctl.

(62) It can be seen that the PG pulse width extends beyond t.sub.p by the delay t.sub.D, and this is accomplished by adding t.sub.D to the PFD reset signal or pulse, which also may be the residual pulse Q.sub.B from the PFD (assuming Q.sub.B is the narrower one relative to Q.sub.A), as explained above. The result is that when the PLL locks and exhibits a typical PFD up and down pulse duty cycle of about 1%, a corresponding PG pulse, and thus the reset current, has a steady-state duty cycle as small as about 3% (subject to the delay t.sub.D), but by one form, by about 10% or less. This reset duration reduction significantly reduces the SR-LF reset power since the reset control signals PG have a pulse width that is about 17 times to 5 times shorter than the conventional pulse width of cp1a and cp2a.

(63) Referring now to FIG. 7, an alternative power gating pulse (PGP) circuit (or rapid reset circuit) 700 may be the same or similar to PGP circuit 500 except with an optional chopper branch or circuit 720. For example, PGP circuit 700 also has a delay circuit 702 with buffers 710, a Boolean circuit 704, and a pulse generator circuit 708 with a flip-flop 712. Otherwise, those elements of the PGP circuit 700 having the same or similar reference numbers (or similar or same names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such. Thus, the components of PGP circuit 700 already described on PGP circuit 500 need not be described again here.

(64) The chopper branch or circuit 720 is provided to cut-off (or chop) the pulse width of PG when the phase error is greater than a threshold, which may be a proportion of a reference cycle, such as when the phase error t.sub.p is more than at least about 180 degrees, exactly 180 degrees, or half the reference cycle. Without the chopping, PG pulse width can exceed half the reference cycle, encroaching the sampling phase of the next reference cycle.

(65) The chopper circuit 720 may have a node Y along the Y line in the Boolean circuit 704 to receive the Y signal at a mux 724 that is arranged to select between Y and a HIGH input 722, which may be tied to supply voltage V.sub.ccpll. The mux 724 also may receive a static selection (SS) signal that selects either HIGH or Y for output Y2. The output Y2 of the mux 724 may be provided to an AND2 gate 726. Due to the timing of the rising edges of the Y pulses as explained above and as input to the mux, this indicates the reference cycle. When Y2 is HIGH, the output of the AND2 falls following the falling edge of the PG. When Y2 is Y, PG is simply passed through as is. The static selector (SS), or just Static, may be programmed to provide a fixed value, and otherwise, the circuit is arranged as shown on FIG. 7.

(66) Referring to FIG. 8, a loop filter 800 may be a sample-reset (SR) loop filter (SR-LF) the same or similar to loop filter 330 (FIG. 3). The SR-LF 800 uses the reset control signals PG and PG' from the PGP circuit 500 or 700 to reduce the reset duration of a proportional path. It is pointed out that those elements of SR-LF 800 having the same or similar reference numbers (or names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such.

(67) In more detail, and in some implementations, sample-reset loop filter 800 has two import ports to receive pumped charge V.sub.ctli with current I.sub.cpi for an integral path 802, and proportional pumped charge V.sub.ctlpp with current I.sub.cpp to a proportional path 804. Both of them receive the charge proportional to the PLL phase error between the reference and feedback clocks. In each reference clock cycle, proportional charges deposited onto C2a or C2b from the previous reference

cycle is reset to V.sub.rst during the present UP/DN event. The output signal V.sub.ctl may be received by a VCO to set or adjust the oscillating frequency of the VCO. In some implementations, a main loop capacitor C.sub.1 is provided on the integral path **802** as shown and is instantiated between V.sub.ccpll, the PLL supply, and V.sub.ctl. In some implementations, the proportional path **804** is based on the sample and reset topology, but a reset circuit **806** may be modified to reset to Vccpll/2, or a different voltage level to improve proportional charge pump current matching between UP and DN.

(68) The proportional path **804** is formed by sample-reset circuitry and comprises sample switches S.sub.1 and S.sub.2 controlled by sampling switch control signals cp1 and cp2, reset switches S.sub.3 and S.sub.4 controlled by switch reset signals cp1a and cp2a, and capacitive devices C.sub.2a and C.sub.2b. Particularly, the capacitors C.sub.2a and C.sub.2b are each coupled to a charge pump via a first node n1, and are coupled in parallel with each other between the first node n1 and a second node n2. The sample switch S.sub.1 is coupled between the capacitor C.sub.2a and the first node n1, while the sample switch S.sub.2 is coupled between the capacitor C.sub.2b and the first node n1. Reset switch S.sub.3 is coupled between the reset circuit **806** and a third node n3 between the capacitor C.sub.2a and the sampling switch S.sub.1, while the reset switch S.sub.4 is coupled between the reset circuit **806** and a fourth node n4 between the capacitor C.sub.2b and the sample switch S.sub.2.

(69) The main loop capacitor C.sub.1 is coupled to the proportional path **804** as shown. V.sub.rst is a reset voltage which is generated by the reset circuit **806** and with a duration corresponding to, or approximately equal to, the shortened pulse widths of reset control signals PG and PG'. The reset signal V.sub.rst may be significantly smaller than Vccpll/2 for example, and in accordance with some implementations herein. Here, switch control signals cp1 and cp2 are derived from the reference clock as explained above. In some implementations, cp1 and cp2 signals are periodic at half of the reference clock rate as shown on timing diagram **600**. In some implementations, cp1 and cp2 are for sampling and open switches S1 and S2 so that capacitors C.sub.2A and alternatively C.sub.2B are ready to take proportional charges.

(70) By one approach, the reset circuit **806** uses a resistor divider with resistors R1 and R2. Specifically, by this approach, reset circuit **806** has transistors MP and MN, and resistors R.sub.1 and R.sub.2, which may be tunable, are coupled together as shown. Transistor MP is controlled by PG' while transistor MN is controlled by PG as obtained from the PGP circuit **500** or **700** for example. The reset control signals PG and PG' can turn the resistor divider off for a time before a pulse of the UP/DN signals. Promptly before and during the reset process, where R.sub.1 and R.sub.2 are connecting (or consuming DC bias current), PG/PG' also is HIGH/LOW.

(71) In some implementations, the node of V.sub.rst may have a decoupling capacitor (not shown) to ground or V.sub.ccpll to mitigate timing mismatch induced Vrst fluctuation that mainly results from the ON time of transistors MN and MP. For example, when PG is HIGH but PG' is not yet LOW due to some timing mismatch (or any other reasons), then the V.sub.ccpll/2 node may get immediately pulled low if there is no de-coupling capacitance connected to this VCC/2 node. With the decoupling cap, the instant voltage fluctuation when **806** begins to turn ON is mitigated, reducing the perturbation from V.sub.rst through C2a or C2b to V.sub.ctl.

(72) In each of the reset circuit examples shown herein, the reset control signal PG and PG' turns the reset current both on and off when the PG pulse first rises to HIGH and then drops back to LOW. Alternatively, reset switch control signals cp1a and cp2a could be provided to reset switches S.sub.3 and S.sub.4 to initiate the reset current (turning ON the reset current), while the falling edge of the switch control signal PG terminates the reset current.

(73) Referring to FIG. 9A, an alternative reset circuit **900** to replace reset circuit **806** may further reduce power consumption and/or circuit area by replacing the passive element resistors with active elements (or devices) in a certain arrangement instead. In this case, reset circuit **900** has a P-channel metal oxide semiconductor (PMOS) diode-connected transistor in conjunction with an N-

channel metal oxide semiconductor (NMOS) diode-connected transistor. Specifically, reset circuit **900** may have a pair of PMOS transistors MP1 and MP2 coupled from drain to source as one branch, and a pair of NMOS transistors MN1 and MN2 coupled from drain to source as another branch with a junction of the two branches at node **902**. The outer transistor MP1 may be coupled to supply voltage V.sub.ccpll at its source, and the outer transistor MN2 may be coupled to ground at its drain.

(74) Inner transistors MP2 and MN1 of the two branches also are coupled from drain to source, and in a stacking arrangement. Node **902** is located along the coupling to provide gate voltage to the two inner transistors MP2 and MN1 as well as provide output reset voltage V.sub.rst to control the two reset switches S.sub.3 and S.sub.4. The outer transistors MP1 and MN2 receive the two reset control pulses PG and PG' as their gate voltage to activate the circuit and output V.sub.rst at pulse widths controlled by PG and PG'. The output V.sub.rst settles to a "trip voltage" of the PMOS pull-up and NMOS pull-down branches, which is not necessarily at Vccpll/2 as with reset circuit **806** (FIG. 8) and other conventional circuits. This arrangement generates a fast-settling reset current. It will be understood that the reset circuit acts as an inverter where the output is short to the input so that both the NMOS and PMOS transistors are acting in saturation mode. It also will be understood that the reset circuit **900** is self-biasing to bias MP2 and MN1 into a saturation mode. Also, the reset circuit **900** has tunable parallel multiples that change the reset current in proportion to the multiples.

(75) The transistors may be sized so that the dc current is about 400 uA, as one example. Also, a decoupling capacitor C3 and supply voltage (or ground) can be provided on this reset circuit **900** as well and as described above with reset circuit **806**. Here, the capacitor C3 may be coupled to the node **902** as well and at the output of the reset circuit providing the V.sub.rst. The supply voltage may be tied to V.sub.ccpll, but could be tied to V.sub.ss instead where a supply voltage pin is connected to a source of the transistor. Otherwise, instead of capacitor C3, two capacitors in series from V.sub.ccpll to V.sub.ss can be used where the junction node in the middle is tied to V.sub.rst.

(76) Referring to FIG. 9B, another alternative reset circuit **950** to replace reset circuit **806** has PMOS transistors QP1 and QP2, and NMOS transistors QN1 and QN2 in a linear mode instead, also similar to having two resistors in series. The reset circuit here also is self-biased, and can have tunable parallel multiples that changes the reset current. Thus, in this example, the four transistors QN1, QN2, QP1, QP2 are coupled drain to source in two branches as shown, and where the two branches are connected at a node **904**, as with reset circuit **900**. The two outer transistors QP1 and QN2 receive the reset control signals PG and PG' respectively at their gates. Different here, the inner transistors QP2 and QN1 each receive the voltage from the respective coupling between inner and outer transistors as their gate voltage. Otherwise, the structure is the same or similar to that of reset circuit **900** and the description of which does not need to be repeated here. The configuration of reset circuit **950** has the advantage of realizing high resistance by operating QP2 and QN1 in the linear mode.

(77) It will be noted that the PLL of various implementations features a modular implementation and therefore can be used as the local clock source or as part of a clock-generation hub. The clock-generation hub provides reference clocks to subsystems from a single platform crystal oscillator through a combination of divisions and distributions, in accordance with some implementations.

(78) For purposes of the implementations herein, the transistors in various circuits, modules, and logic blocks are metal oxide semiconductor (MOS) transistors, which include drain, source, gate, and bulk terminals. The transistors also include Tri-Gate and FinFET transistors, Gate All Around Cylindrical Transistors, Tunneling FET (TFET), Square Wire, or Rectangular Ribbon Transistors or other devices implementing transistor functionality like carbon nano tubes or spintronic devices. MOSFET symmetrical source and drain terminals i.e., are identical terminals and are interchangeably used here. A TFET device, on the other hand, has asymmetric Source and Drain terminals. It will be appreciated that other transistors, for example, Bi-polar junction transistors—

BJT PNP/NPN, BiCMOS, CMOS, eFET, etc., may be used without departing from the scope of the disclosure.

(79) Referring to FIG. **10**, a computer system or computing device **1000** (also referred to as device **1000**) has a number of elements that may have, or may be operated by, a PLL (or DLL) such as PLL **200** (FIG. **2**) or **300** (FIG. **3**) in accordance with some implementations herein. For example, the PLL **200** has a power gating pulse circuit (or just pulse circuit) **212** (or **500** of FIG. **5**) that provides reset control pulses to a reset circuit **214** (or **806**, **900**, or **950** of FIGS. **8-9B** respectively). The pulse circuit **500** generates a time delay to add to a signal related to the phase error or up-dn signals of a phase frequency detector (PFD) **202** or **400** (FIG. **4**). The time delay as well as initial switch control circuits input to the pulse circuit are used to generate short pulse widths for the reset control pulses before providing the reset control pulses to the reset circuit to control the proportional path of the loop filter as described above. One or more of the processors on device **1000** as well as the other elements forming device **1000** may have or operate the PLL **200** or **300** as described above. It is pointed out that those elements of FIG. **10** having the same reference numbers (or names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such.

(80) In some implementations, device **1000** represents an appropriate computing device, such as a computing tablet, a mobile phone or smart-phone, a laptop, a desktop, an Internet-of-Things (JOT) device, a server, a wearable device, a set-top box, a wireless-enabled e-reader, or the like. It will be understood that certain components are shown generally, and not all components of such a device are shown in device **1000**.

(81) In an example, the device **1000** comprises a SoC (System-on-Chip) **1001**. An example boundary of the SOC **1001** is illustrated using dotted lines in FIG. **10**, with some example components being illustrated to be included within SOC **1001**—however, SOC **1001** may include any appropriate components of device **1000**.

(82) In some implementations, device **1000** includes processor **1004**. Processor **1004** can include one or more physical devices, such as microprocessors, application processors, microcontrollers, programmable logic devices, processing cores, or other processing means. The processing operations performed by processor **1004** include the execution of an operating platform or operating system on which applications and/or device functions are executed. The processing operations include operations related to I/O (input/output) with a human user or with other devices, operations related to power management, operations related to connecting computing device **1000** to another device, and/or the like. The processing operations may also include operations related to audio I/O and/or display I/O.

(83) In some implementations, processor **1004** includes multiple processing cores (also referred to as cores) **1008a**, **1008b**, **1008c**. Although merely three cores **1008a**, **1008b**, **1008c** are illustrated in FIG. **10**, the processor **1004** may include any other appropriate number of processing cores, e.g., tens, or even hundreds of processing cores. Processor cores **1008a**, **1008b**, **1008c** may be implemented on a single integrated circuit (IC) chip. Moreover, the chip may include one or more shared and/or private caches, buses or interconnections, graphics and/or memory controllers, or other components.

(84) In some implementations, processor **1004** includes cache **1006**. In an example, sections of cache **1006** may be dedicated to individual cores **1008** (e.g., a first section of cache **1006** dedicated to core **1008a**, a second section of cache **1006** dedicated to core **1008b**, and so on). In an example, one or more sections of cache **1006** may be shared among two or more of cores **1008**. Cache **1006** may be split in different levels, e.g., level 1 (L1) cache, level 2 (L2) cache, level 3 (L3) cache, etc.

(85) In some implementations, a given processor core (e.g., core **1008a**) may include a fetch unit to fetch instructions (including instructions with conditional branches) for execution by the core **1008a**. The instructions may be fetched from any storage devices such as the memory **1030**. Processor core **1008a** may also include a decode unit to decode the fetched instruction. For

example, the decode unit may decode the fetched instruction into a plurality of micro-operations. Processor core **1008a** may include a schedule unit to perform various operations associated with storing decoded instructions. For example, the schedule unit may hold data from the decode unit until the instructions are ready for dispatch, e.g., until all source values of a decoded instruction become available. In one implementation, the schedule unit may schedule and/or issue (or dispatch) decoded instructions to an execution unit for execution.

(86) The execution unit may execute the dispatched instructions after they are decoded (e.g., by the decode unit) and dispatched (e.g., by the schedule unit). In an implementation, the execution unit may include more than one execution unit (such as an imaging computational unit, a graphics computational unit, a general-purpose computational unit, etc.). The execution unit may also perform various arithmetic operations such as addition, subtraction, multiplication, and/or division, and may include one or more arithmetic logic units (ALUs). In an implementation, a co-processor (not shown) may perform various arithmetic operations in conjunction with the execution unit.

(87) Further, an execution unit may execute instructions out-of-order. Hence, processor core **1008a** (for example) may be an out-of-order processor core in one implementation. Processor core **1008a** may also include a retirement unit. The retirement unit may retire executed instructions after they are committed. In an implementation, retirement of the executed instructions may result in processor state being committed from the execution of the instructions, physical registers used by the instructions being de-allocated, etc. The processor core **1008a** may also include a bus unit to enable communication between components of the processor core **1008a** and other components via one or more buses. Processor core **1008a** may also include one or more registers to store data accessed by various components of the core **1008a** (such as values related to assigned app priorities and/or sub-system states (modes) association).

(88) In some implementations, device **1000** comprises connectivity circuitries **1031**. For example, connectivity circuitries **1031** includes hardware devices (e.g., wireless and/or wired connectors and communication hardware) and/or software components (e.g., drivers, protocol stacks), e.g., to enable device **1000** to communicate with external devices. Device **1000** may be separate from the external devices, such as other computing devices, wireless access points or base stations, etc.

(89) In an example, connectivity circuitries **1031** may include multiple different types of connectivity. To generalize, the connectivity circuitries **1031** may include cellular connectivity circuitries, wireless connectivity circuitries, etc. Cellular connectivity circuitries of connectivity circuitries **1031** refers generally to cellular network connectivity provided by wireless carriers, such as provided via GSM (global system for mobile communications) or variations or derivatives, CDMA (code division multiple access) or variations or derivatives, TDM (time division multiplexing) or variations or derivatives, 3rd Generation Partnership Project (3GPP) Universal Mobile Telecommunications Systems (UMTS) system or variations or derivatives, 3GPP Long-Term Evolution (LTE) system or variations or derivatives, 3GPP LTE-Advanced (LTE-A) system or variations or derivatives, Fifth Generation (5G) wireless system or variations or derivatives, 5G mobile networks system or variations or derivatives, 5G New Radio (NR) system or variations or derivatives, or other cellular service standards. Wireless connectivity circuitries (or wireless interface) of the connectivity circuitries **1031** refers to wireless connectivity that is not cellular, and can include personal area networks (such as Bluetooth, Near Field, etc.), local area networks (such as Wi-Fi), and/or wide area networks (such as WiMax), and/or other wireless communication. In an example, connectivity circuitries **1031** may include a network interface, such as a wired or wireless interface, e.g., so that a system implementation may be incorporated into a wireless device, for example, cell phone or personal digital assistant.

(90) In some implementations, device **1000** comprises control hub **1032**, which represents hardware devices and/or software components related to interaction with one or more I/O devices. For example, processor **1004** may communicate with one or more of display **1022**, one or more

peripheral devices **1024**, storage devices **1028**, one or more other external devices **1029**, etc., via control hub **1032**. Control hub **1032** may be a chipset, a Platform Control Hub (PCH), and/or the like.

(91) For example, control hub **1032** illustrates one or more connection points for additional devices that connect to device **1000**, e.g., through which a user might interact with the system. For example, devices (e.g., devices **1029**) that can be attached to device **1000** include microphone devices, speaker or stereo systems, audio devices, video systems or other display devices, keyboard or keypad devices, or other I/O devices for use with specific applications such as card readers or other devices.

(92) As mentioned above, control hub **1032** can interact with audio devices, display **1022**, etc. For example, input through a microphone or other audio device can provide input or commands for one or more applications or functions of device **1000**. Additionally, audio output can be provided instead of, or in addition to display output. In another example, if display **1022** includes a touch screen, display **1022** also acts as an input device, which can be at least partially managed by control hub **1032**. There can also be additional buttons or switches on computing device **1000** to provide I/O functions managed by control hub **1032**. In one implementation, control hub **1032** manages devices such as accelerometers, cameras, light sensors or other environmental sensors, or other hardware that can be included in device **1000**. The input can be part of direct user interaction, as well as providing environmental input to the system to influence its operations (such as filtering for noise, adjusting displays for brightness detection, applying a flash for a camera, or other features).

(93) In some implementations, control hub **1032** may couple to various devices using any appropriate communication protocol, e.g., PCIe (Peripheral Component Interconnect Express), USB (Universal Serial Bus), Thunderbolt, High Definition Multimedia Interface (HDMI), Firewire, etc.

(94) In some implementations, display **1022** represents hardware (e.g., display devices) and software (e.g., drivers) components that provide a visual and/or tactile display for a user to interact with device **1000**. Display **1022** may include a display interface, a display screen, and/or hardware device used to provide a display to a user. In some implementations, display **1022** includes a touch screen (or touch pad) device that provides both output and input to a user. In an example, display **1022** may communicate directly with the processor **1004**. Display **1022** can be one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). In one implementation display **1022** can be a head mounted display (HMD) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

(95) In some implementations and although not illustrated in the figure, in addition to (or instead of) processor **1004**, device **1000** may include Graphics Processing Unit (GPU) comprising one or more graphics processing cores, which may control one or more aspects of displaying contents on display **1022**.

(96) Control hub **1032** (or platform controller hub) may include hardware interfaces and connectors, as well as software components (e.g., drivers, protocol stacks) to make peripheral connections, e.g., to peripheral devices **1024**.

(97) It will be understood that device **1000** could both be a peripheral device to other computing devices, as well as have peripheral devices connected to it. Device **1000** may have a “docking” connector to connect to other computing devices for purposes such as managing (e.g., downloading and/or uploading, changing, synchronizing) content on device **1000**. Additionally, a docking connector can allow device **1000** to connect to certain peripherals that allow computing device **1000** to control content output, for example, to audiovisual or other systems.

(98) In addition to a proprietary docking connector or other proprietary connection hardware, device **1000** can make peripheral connections via common or standards-based connectors. Common types can include a Universal Serial Bus (USB) connector (which can include any of a

number of different hardware interfaces), DisplayPort including MiniDisplayPort (MDP), High Definition Multimedia Interface (HDMI), Firewire, or other types.

(99) In some implementations, connectivity circuitries **1031** may be coupled to control hub **1032**, e.g., in addition to, or instead of, being coupled directly to the processor **1004**. In some implementations, display **1022** may be coupled to control hub **1032**, e.g., in addition to, or instead of, being coupled directly to processor **1004**.

(100) In some implementations, device **1000** comprises memory **1030** coupled to processor **1004** via memory interface **1034**. Memory **1030** includes memory devices for storing information in device **1000**. Memory can include nonvolatile (state does not change if power to the memory device is interrupted) and/or volatile (state is indeterminate if power to the memory device is interrupted) memory devices. Memory device **1030** can be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. In one implementation, memory **1030** can operate as system memory for device **1000**, to store data and instructions for use when the one or more processors **1004** executes an application or process. Memory **1030** can store application data, user data, music, photos, documents, or other data, as well as system data (whether long-term or temporary) related to the execution of the applications and functions of device **1000**.

(101) Elements of various implementations and examples are also provided as a machine-readable medium (e.g., memory **1030**) for storing the computer-executable instructions (e.g., instructions to implement any other processes discussed herein). The machine-readable medium (e.g., memory **1030**) may include, but is not limited to, flash memory, optical disks, CD-ROMs, DVD ROMs, RAMs, EPROMs, EEPROMs, magnetic or optical cards, phase change memory (PCM), or other types of machine-readable media suitable for storing electronic or computer-executable instructions. For example, implementations of the disclosure may be downloaded as a computer program (e.g., BIOS) which may be transferred from a remote computer (e.g., a server) to a requesting computer (e.g., a client) by way of data signals via a communication link (e.g., a modem or network connection).

(102) In some implementations, device **1000** comprises temperature measurement circuitries **1040**, e.g., for measuring temperature of various components of device **1000**. In an example, temperature measurement circuitries **1040** may be embedded, or coupled or attached to various components, whose temperature are to be measured and monitored. For example, temperature measurement circuitries **1040** may measure temperature of (or within) one or more of cores **1008a**, **1008b**, **1008c**, voltage regulator **1014**, memory **1030**, a mother-board of SOC **1001**, and/or any appropriate component of device **1000**.

(103) In some implementations, device **1000** comprises power measurement circuitries **1042**, e.g., for measuring power consumed by one or more components of the device **1000**. In an example, in addition to, or instead of, measuring power, the power measurement circuitries **1042** may measure voltage and/or current. In an example, the power measurement circuitries **1042** may be embedded, or coupled or attached to various components, whose power, voltage, and/or current consumption are to be measured and monitored. For example, power measurement circuitries **1042** may measure power, current and/or voltage supplied by one or more voltage regulators **1014**, power supplied to SOC **1001**, power supplied to device **1000**, power consumed by processor **1004** (or any other component) of device **1000**, etc.

(104) In some implementations, device **1000** comprises one or more voltage regulator circuitries, generally referred to as voltage regulator (VR) **1014**. VR **1014** generates signals at appropriate voltage levels, which may be supplied to operate any appropriate components of the device **1000**. Merely as an example, VR **1014** is illustrated to be supplying signals to processor **1004** of device **1000**. In some implementations, VR **1014** receives one or more Voltage Identification (VID) signals, and generates the voltage signal at an appropriate level, based on the VID signals. Various

type of VRs may be utilized for the VR **1014**. For example, VR **1014** may include a “buck” VR, “boost” VR, a combination of buck and boost VRs, low dropout (LDO) regulators, switching DC-DC regulators, etc. Buck VR is generally used in power delivery applications in which an input voltage needs to be transformed to an output voltage in a ratio that is smaller than unity. Boost VR is generally used in power delivery applications in which an input voltage needs to be transformed to an output voltage in a ratio that is larger than unity. In some implementations, each processor core has its own VR which is controlled by PCU **1010a/b** and/or PMIC **1012**. In some implementations, each core has a network of distributed LDOs to provide efficient control for power management. The LDOs can be digital, analog, or a combination of digital or analog LDOs.

(105) In some implementations, device **1000** comprises one or more clock generator circuitries, generally referred to as clock generator **1016**. Clock generator **1016** generates clock signals at appropriate frequency levels, which may be supplied to any appropriate components of device **1000**. Merely as an example, clock generator **1016** is illustrated to be supplying clock signals to processor **1004** of device **1000**. In some implementations, clock generator **1016** receives one or more Frequency Identification (FID) signals, and generates the clock signals at an appropriate frequency, based on the FID signals.

(106) In some implementations, device **1000** comprises battery **1018** supplying power to various components of device **1000**. Merely as an example, battery **1018** is illustrated to be supplying power to processor **1004**. Although not illustrated in the figures, device **1000** may comprise a charging circuitry, e.g., to recharge the battery, based on Alternating Current (AC) power supply received from an AC adapter.

(107) In some implementations, device **1000** comprises Power Control Unit (PCU) **1010** (also referred to as Power Management Unit (PMU), Power Controller, etc.). In an example, some sections of PCU **1010** may be implemented by one or more processing cores **1008**, and these sections of PCU **1010** are symbolically illustrated using a dotted box and labelled PCU **1010a**. In an example, some other sections of PCU **1010** may be implemented outside the processing cores **1008**, and these sections of PCU **1010** are symbolically illustrated using a dotted box and labelled as PCU **1010b**. PCU **1010** may implement various power management operations for device **1000**. PCU **1010** may include hardware interfaces, hardware circuitries, connectors, registers, etc., as well as software components (e.g., drivers, protocol stacks), to implement various power management operations for device **1000**.

(108) In some implementations, device **1000** comprises Power Management Integrated Circuit (PMIC) **1012**, e.g., to implement various power management operations for device **1000**. In some implementations, PMIC **1012** is a Reconfigurable Power Management ICs (RPMICs) and/or an IMVP (Intel® Mobile Voltage Positioning). In an example, the PMIC is within an IC chip separate from processor **1004**. The may implement various power management operations for device **1000**. PMIC **1012** may include hardware interfaces, hardware circuitries, connectors, registers, etc., as well as software components (e.g., drivers, protocol stacks), to implement various power management operations for device **1000**.

(109) In an example, device **1000** comprises one or both PCU **1010** or PMIC **1012**. In an example, any one of PCU **1010** or PMIC **1012** may be absent in device **1000**, and hence, these components are illustrated using dotted lines.

(110) Various power management operations of device **1000** may be performed by PCU **1010**, by PMIC **1012**, or by a combination of PCU **1010** and PMIC **1012**. For example, PCU **1010** and/or PMIC **1012** may select a power state (e.g., P-state) for various components of device **1000**. For example, PCU **1010** and/or PMIC **1012** may select a power state (e.g., in accordance with the ACPI (Advanced Configuration and Power Interface) specification) for various components of device **1000**. Merely as an example, PCU **1010** and/or PMIC **1012** may cause various components of the device **1000** to transition to a sleep state, to an active state, to an appropriate C state (e.g., C0 state, or another appropriate C state, in accordance with the ACPI specification), etc. In an example, PCU

1010 and/or **PMIC 1012** may control a voltage output by **VR 1014** and/or a frequency of a clock signal output by the clock generator, e.g., by outputting the **VID** signal and/or the **FID** signal, respectively. In an example, **PCU 1010** and/or **PMIC 1012** may control battery power usage, charging of battery **1018**, and features related to power saving operation.

(111) The clock generator **1016** can comprise the phase locked loop (PLL) described herein, frequency locked loop (FLL), or any suitable clock source. In some implementations, each core of processor **1004** has its own clock source. As such, each core can operate at a frequency independent of the frequency of operation of the other core. In some implementations, **PCU 1010** and/or **PMIC 1012** performs adaptive or dynamic frequency scaling or adjustment. For example, clock frequency of a processor core can be increased if the core is not operating at its maximum power consumption threshold or limit. In some implementations, **PCU 1010** and/or **PMIC 1012** determines the operating condition of each core of a processor, and opportunistically adjusts frequency and/or power supply voltage of that core without the core clocking source (e.g., PLL of that core) losing lock when the **PCU 1010** and/or **PMIC 1012** determines that the core is operating below a target performance level. For example, if a core is drawing current from a power supply rail less than a total current allocated for that core or processor **1004**, then **PCU 1010** and/or **PMIC 1012** can temporarily increase the power draw for that core or processor **1004** (e.g., by increasing clock frequency and/or power supply voltage level) so that the core or processor **1004** can perform at a higher performance level. As such, voltage and/or frequency can be increased temporally for processor **1004** without violating product reliability.

(112) In an example, **PCU 1010** and/or **PMIC 1012** may perform power management operations, e.g., based at least in part on receiving measurements from power measurement circuitries **1042**, temperature measurement circuitries **1040**, charge level of battery **1018**, and/or any other appropriate information that may be used for power management. To that end, **PMIC 1012** is communicatively coupled to one or more sensors to sense/detect various values/variations in one or more factors having an effect on power/thermal behavior of the system/platform. Examples of the one or more factors include electrical current, voltage droop, temperature, operating frequency, operating voltage, power consumption, inter-core communication activity, etc. One or more of these sensors may be provided in physical proximity (and/or thermal contact/coupling) with one or more components or logic/IP blocks of a computing system. Additionally, sensor(s) may be directly coupled to **PCU 1010** and/or **PMIC 1012** in at least one implementation to allow **PCU 1010** and/or **PMIC 1012** to manage processor core energy at least in part based on value(s) detected by one or more of the sensors.

(113) Also illustrated is an example software stack of device **1000** (although not all elements of the software stack are illustrated). Merely as an example, processors **1004** may execute application programs **1050**, Operating System **1052**, one or more Power Management (PM) specific application programs (e.g., generically referred to as PM applications **1058**), and/or the like. PM applications **1058** may also be executed by the **PCU 1010** and/or **PMIC 1012**. OS **1052** may also include one or more PM applications **1056a**, **1056b**, **1056c**. The OS **1052** may also include various drivers **1054a**, **1054b**, **1054c**, etc., some of which may be specific for power management purposes. In some implementations, device **1000** may further comprise a Basic Input/Output System (BIOS) **1020**. BIOS **1020** may communicate with OS **1052** (e.g., via one or more drivers **1054**), communicate with processors **1004**, etc.

(114) For example, one or more of PM applications **1058**, **1056**, drivers **1054**, BIOS **1020**, etc. may be used to implement power management specific tasks, e.g., to control voltage and/or frequency of various components of device **1000**, to control wake-up state, sleep state, and/or any other appropriate power state of various components of device **1000**, control battery power usage, charging of the battery **1018**, features related to power saving operation, etc.

(115) Furthermore, the particular features, structures, functions, or characteristics may be combined in any suitable manner in one or more implementations. For example, a first implementation may

be combined with a second implementation anywhere the particular features, structures, functions, or characteristics associated with the two implementations are not mutually exclusive.

(116) While the disclosure has been described in conjunction with specific implementations thereof, many alternatives, modifications and variations of such implementations will be apparent to those of ordinary skill in the art in light of the foregoing description. For example, other memory architectures e.g., Dynamic RAM (DRAM) may use the implementations discussed. The implementations of the disclosure are intended to embrace all such alternatives, modifications, and variations as to fall within the broad scope of the appended claims.

(117) In addition, well known power/ground connections to integrated circuit (IC) chips and other components may or may not be shown within the presented figures, for simplicity of illustration and discussion, and so as not to obscure the disclosure. Further, arrangements may be shown in block diagram form in order to avoid obscuring the disclosure, and also in view of the fact that specifics with respect to implementation of such block diagram arrangements are highly dependent upon the platform within which the present disclosure is to be implemented (i.e., such specifics should be well within purview of one skilled in the art). Where specific details (e.g., circuits) are set forth in order to describe example implementations of the disclosure, it should be apparent to one skilled in the art that the disclosure can be practiced without, or with variation of, these specific details. The description is thus to be regarded as illustrative instead of limiting.

(118) The following examples pertain to further implementations. Specifics in the examples may be used anywhere in one or more implementations. All optional features of the apparatus described herein may also be implemented with respect to a method or process.

(119) Example 1 is an apparatus which comprises: first circuitry to use a first reset signal to generate a second reset signal, wherein the first reset signal is to be further provided to reset a phase frequency detector circuit; second circuitry coupled to the first circuitry and comprising: one or more Boolean operators to generate a first signal based on first and second switch control clocks, and a flip-flop circuit configured to output a second signal and a third signal complimentary to the second signal, wherein the second signal is based on the first signal and second reset signal; and wherein the second circuitry is to provide the second and third signal to a reset circuit of a loop filter circuit having a first switch and a second switch each coupled between the reset circuit and a different respective capacitor of the loop filter circuit; and wherein the reset circuit provides a first switch control signal and a second switch control signal respectively to the first and second switches.

(120) Example 2 includes all features of example 1, and wherein the pulse width of the second and third signals corresponds to a duration the reset circuit consumes power performing a reset of the capacitors.

(121) Example 3 includes all features of example 1 or 2, and wherein the first reset signal is based on a phase error between a reference clock and a feedback clock.

(122) Example 4 includes all features of any of example 1-3, and wherein the second reset signal is formed by applying a delay to the first reset signal.

(123) Example 5 includes all features of example 4, and wherein a tunable duration of the delay is pre-determined and results in adding a fixed duration delay to a duration of a phase error between a reference clock and a feedback clock used to form the first reset signal.

(124) Example 6 includes all features of example 4, wherein a tunable duration of the delay is automatically adjusted so that varying delay durations are added to the duration of a phase error between a reference clock and a feedback clock used to form the first reset signal.

(125) Example 7 includes all features of any of examples 1-6, wherein the first capacitor and the second capacitor are each coupled to a charge pump via a first node; and the first capacitor and the second capacitor are coupled in parallel with each other between the first node and a second node.

(126) Example 8 includes all features of example 7, wherein the first switch is coupled to the first capacitor via a third node, wherein a third switch of the loop filter circuit is coupled between the

first node and the third node; and the second switch is coupled to the second capacitor via a fourth node, wherein a fourth switch of the loop filter circuit is coupled between the first node and the fourth node.

(127) Example 9 includes all features of example 1, and wherein the power gating pulse circuit comprises a chopper branch to cut-off a pulse width of the pulses of the reset control signals when the phase error extends longer than a threshold and relative to the reference clock.

(128) Example 10 includes an apparatus that comprises: a loop filter to receive input and to output a control signal based on the input, the loop filter comprising capacitors coupled to the input and a reset circuit to reset the capacitors; and a power gating pulse circuit comprising pulse generator circuitry configured to adaptively generate complimentary reset control signals to control the reset circuit, wherein a pulse width of the reset control signals is based on both a phase error between a reference clock and a feedback clock and switch control signals based on the reference clock.

(129) Example 11 includes all features of example 10, wherein the pulse width of the reset control signal corresponds to a duration the reset circuit consumes power performing a reset of the capacitors.

(130) Example 12 includes all features of example 10 or 11, wherein the power gating pulse circuit comprises a delay circuit arranged to apply a delay to a first signal depending on the phase error and generated by a phase frequency detector (PFD), and to generate a second signal, and wherein the pulse generator circuitry uses the second signal to determine the pulse widths of the reset control signals.

(131) Example 13 includes all features of any of examples 10-12, wherein the pulse generator circuit comprises a flip-flop circuitry, and wherein switch control signals are capacitor reset clocks with a pulse width fixed at a proportion of a reference clock and input to the flip-flop circuitry.

(132) Example 14 includes all features of any of examples 10-13, wherein the pulse generator circuit comprises a flip-flop circuitry, and wherein the flip-flop circuitry comprises a D-type flip-flop to be reset by the second signal, receives the switch control signals as input, and outputs the reset control signals.

(133) Example 15 includes all features of examples 10-14, wherein the power gating pulse circuit comprises a Boolean logic circuit that receives the switch control signals, and wherein the pulse generator circuit comprises a flip-flop that receives an input generated by the Boolean logic and outputs the reset control signals.

(134) Example 16 includes all features of example 15, and wherein the Boolean logic comprises an OR gate to receive alternating switch control signals that have pulse widths fixed to a proportion of the reference clock.

(135) Example 17 includes all features of any of examples 10-16, and wherein the switch control signals have pulse widths fixed to a proportion of the reference clock, wherein the pulse generator circuit is a flip-flop circuitry, and wherein the flip-flop circuitry uses the switch control signals to set the start of pulses on the reset control signals, wherein the flip-flop circuitry receives a flip-flop reset signal to set the end of the pulses of the reset control signals, and wherein the flip-flop reset signal being based on detection of the phase error.

(136) Example 18 includes all features of any of examples 10-17, wherein the reset circuit does not have resistors so that the reset circuit resets the capacitors without using resistors.

(137) Example 19 is a system that comprises: memory; processor circuitry forming a processor coupled to the memory, the processor having a loop filter comprising: a first capacitor and a second capacitor each coupled to a charge pump via a first node, wherein the first capacitor and the second capacitor are coupled in parallel with each other between the first node and a second node; a first switch coupled between the first capacitor and the first node; a second switch coupled between the second capacitor and the first node; a reset circuit to receive a first signal and a second signal which is complementary to the first signal, wherein pulse widths of the first and second signals depend on a phase error between clocks used to provide charge from the charge pump; a third switch coupled

between the reset circuit and a third node between the first capacitor and the first switch; and a fourth switch coupled between the reset circuit and a fourth node between the second capacitor and the second switch; wherein the reset circuit is configured to provide control signals to the third and fourth switches with timing that depends on the pulse widths; and a wireless interface to allow the processor to communicate with another device.

(138) Example 20 includes all features of claim **19**, and wherein the first and second signals are generated by applying a delay to a phase frequency detector reset signal associated with the phase error to generate a flip-flop reset signal, and resetting a flip-flop of a power gating pulse circuit with the flip-flop reset signal.

(139) Example 21 includes all features of claim **20**, and wherein the first and second signals are also based on alternating switch control signals based on a reference clock used to detect the phase error.

(140) Example 22 includes all features of any of examples 19-21, and wherein the reset circuit comprises an active element voltage divider with active elements coupled in series and without resistors between the active elements.

(141) Example 23 includes all features of any of examples 19-22, and wherein the reset circuit comprises at least four transistors coupled in series including two inner transistors and two outer transistors, wherein the two inner transistors having gates that receive current from a source-drain coupling between two of the transistors.

(142) Example 24 includes all features of any of examples 19-23, and wherein the reset circuit comprises cascading transistors with a P-type diode-connected transistor in conjunction with an N-type diode-connected transistor, and without the use of resistors between the transistors.

(143) Example 25 includes all features of any of examples 19-23, and wherein the reset circuit comprises a P-type transistor and an N-type transistor in saturation mode, and without the use of resistors between the transistors.

Claims

1. An apparatus comprising: first circuitry to use a first reset signal to generate a second reset signal, wherein the first reset signal is to be further provided to reset a phase frequency detector circuit; second circuitry coupled to the first circuitry and comprising: a Boolean logic circuit to generate a first signal based on first and second reset control signals, and a flip-flop circuit configured to output a second signal and a third signal complimentary to the second signal, wherein the second signal is based on the first signal and second reset signal; and wherein the second circuitry is to provide the second and third signal to a reset circuit of a loop filter circuit having a first switch and a second switch each coupled between the reset circuit and a different respective capacitor of first and second capacitors of the loop filter circuit; and wherein the reset circuit provides the first and second reset control signals respectively to the first and second switches.
2. The apparatus of claim 1, wherein the pulse width of the second and third signals corresponds to a duration the reset circuit consumes power performing a reset of the capacitors.
3. The apparatus of claim 1, wherein the first reset signal is based on a phase error between a reference clock and a feedback clock.
4. The apparatus of claim 3, wherein the flip-flop circuit comprises a chopper branch to cut-off a pulse width of the second and third signals when the phase error extends longer than a threshold and relative to the reference clock.
5. The apparatus of claim 1, wherein the second reset signal is formed by applying a delay to the first reset signal.
6. The apparatus of claim 5, wherein a tunable duration of the delay is pre-determined and results in adding a fixed duration delay to a duration of a phase error between a reference clock and a feedback clock used to form the first reset signal.

7. The apparatus of claim 5, wherein a tunable duration of the delay is automatically adjusted so that varying delay durations are added to the duration of a phase error between a reference clock and a feedback clock used to form the first reset signal.
8. The apparatus of claim 1, wherein the first capacitor and the second capacitor are each coupled to a charge pump via a first node; and the first capacitor and the second capacitor are coupled in parallel with each other between the first node and a second node.
9. The apparatus of claim 8, wherein the first switch is coupled to the first capacitor via a third node, wherein a third switch of the loop filter circuit is coupled between the first node and the third node; and the second switch is coupled to the second capacitor via a fourth node, wherein a fourth switch of the loop filter circuit is coupled between the first node and the fourth node.
10. An apparatus comprising: a loop filter to receive input and to output a control signal based on the input, the loop filter comprising capacitors coupled to the input and a reset circuit to reset the capacitors; and a power gating pulse circuit comprising pulse generator circuitry configured to adaptively generate complimentary reset control signals to control the reset circuit, wherein a pulse width of the reset control signals is based on both a phase error between a reference clock and a feedback clock and switch control signals based on the reference clock.
11. The apparatus of claim 10, wherein the pulse width of the reset control signal corresponds to a duration the reset circuit consumes power performing a reset of the capacitors.
12. The apparatus of claim 10, wherein the power gating pulse circuit comprises a delay circuit arranged to apply a delay to a first signal depending on the phase error and generated by a phase frequency detector (PFD), and to generate a second signal, and wherein the pulse generator circuitry uses the second signal to determine the pulse widths of the reset control signals.
13. The apparatus of claim 10, wherein the pulse generator circuit comprises a flip-flop circuitry, and wherein the switch control signals are capacitor reset clocks with a pulse width fixed at a proportion of a reference clock and input to the flip-flop circuitry.
14. The apparatus of claim 10, wherein the pulse generator circuit comprises a flip-flop circuitry, and wherein the flip-flop circuitry comprises a D-type flip-flop to receive the switch control signals as input, and outputs the reset control signals.
15. The apparatus of claim 10, wherein the power gating pulse circuit comprises a Boolean logic circuit that receives the switch control signals, and wherein the pulse generator circuit comprises a flip-flop that receives an input generated by the Boolean logic and outputs the reset control signals.
16. The apparatus of claim 15, wherein the Boolean logic comprises an OR gate to receive alternating switch control signals that have pulse widths fixed to a proportion of the reference clock.
17. The apparatus of claim 10 wherein the switch control signals have pulse widths fixed to a proportion of the reference clock, wherein the pulse generator circuit is a flip-flop circuitry, and wherein the flip-flop circuitry uses the switch control signals to set the start of pulses on the reset control signals, wherein the flip-flop circuitry receives a flip-flop reset signal to set the end of the pulses of the reset control signals, and wherein the flip-flop reset signal being based on detection of the phase error.
18. The apparatus of claim 10, wherein the reset circuit does not have resistors so that the reset circuit resets the capacitors without using resistors.
19. A system comprising: memory; processor circuitry forming a processor coupled to the memory, the processor having a loop filter comprising: a first capacitor and a second capacitor each coupled to a charge pump via a first node, wherein the first capacitor and the second capacitor are coupled in parallel with each other between the first node and a second node; a first switch coupled between the first capacitor and the first node; a second switch coupled between the second capacitor and the first node; a reset circuit to receive a first signal and a second signal which is complementary to the first signal, wherein pulse widths of the first and second signals depend on a phase error between clocks used to provide charge from the charge pump; a third switch coupled between the reset

circuit and a third node between the first capacitor and the first switch; and a fourth switch coupled between the reset circuit and a fourth node between the second capacitor and the second switch; wherein the reset circuit is configured to provide control signals to the third and fourth switches; and a wireless interface to allow the processor to communicate with another device.

20. The system of claim 19 wherein the first and second signals are generated by applying a delay to a phase frequency detector reset signal associated with the phase error to generate a flip-flop reset signal, and resetting a flip-flop of a power gating pulse circuit with the flip-flop reset signal.

21. The system of claim 20 wherein the first and second signals are also based on alternating switch control signals based on a reference clock used to detect the phase error.

22. The system of claim 19 wherein the reset circuit comprises an active element voltage divider with active elements coupled in series and without resistors between the active elements.

23. The system of claim 19 wherein the reset circuit comprises at least four transistors coupled in series including two inner transistors and two outer transistors, wherein the two inner transistors having gates that receive current from a source-drain coupling between two of the transistors.

24. The system of claim 19 wherein the reset circuit comprises cascading transistors with a P-type diode-connected transistor in conjunction with an N-type diode-connected transistor, and without the use of resistors between the transistors.

25. The system of claim 19 wherein the reset circuit comprises a P-type transistor and an N-type transistor in saturation mode, and without the use of resistors between the transistors.
